ReCLLaMA: Appendix

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A Formal Reasoning Rules

This appendix provides additional details for the symbolic reasoning engine used in ReCLLaMA.

A.1 Truth Values

Each statement carries:

$$\langle f, c \rangle,$$

where:

- f: frequency / empirical support,
- c: confidence / certainty.

A.2 Deduction

$$\{M \to P\langle f_1, c_1 \rangle,\ S \to M\langle f_2, c_2 \rangle\} \vdash S \to P\langle f_1 f_2,\ f_1 f_2 c_1 c_2 \rangle. \tag{1}$$

Example:

$$\textit{protein} \to \textit{gene}, \quad \textit{patient} \to \textit{protein} \Rightarrow \textit{patient} \to \textit{gene}. \tag{2}$$

A.3 Abduction

$$\{P \to M,\ S \to M\} \vdash S \to P. \tag{3}$$

Example:

$$\textit{gene} \to \textit{disease}, \quad \textit{patient} \to \textit{disease} \Rightarrow \textit{patient} \to \textit{gene}. \tag{4}$$

A.4 Revision

$$S \to P\langle 1.0, 0.80\rangle, \quad S \to P\langle 1.0, 0.80\rangle \Rightarrow S \to P\langle 1.0, 0.89\rangle. \tag{5}$$

A.5 Choice

Given candidate diagnoses:

$$\mathcal{D} = \{(d_1, c_1), \ldots, (d_n, c_n)\}, \tag{6}$$

the system returns:

$$TopK(\mathcal{D}). \tag{7}$$

A.6 Extended Biomedical Example

Suppose:

$$(patient, \texttt{PROTEIN:6548}), \tag{8}$$
$$(\texttt{PROTEIN:6548}, \texttt{GENE:32979}), \tag{9}$$
$$(\texttt{GENE:32979}, \texttt{75839}). \tag{10}$$

Then deduction yields:

$$(patient, \texttt{GENE:32979}) \to has_gene, \tag{11}$$

and then:

$$(patient, \texttt{75839}) \to potential_disease. \tag{12}$$

This provides a complete symbolic path from patient evidence to diagnosis.

A.7 Scalability

Current experiments use shallow two-hop reasoning. However, the framework naturally extends to deeper reasoning chains over larger biomedical graphs while preserving transparency and confidence calibration.

B Human-designed Prompts

B.1 Prompt Design for Diagnostic Entity Extraction and Interpretation

To support automated clinical reasoning, we design two modular prompts targeting distinct stages of the pipeline: (1) diagnostic entity extraction and (2) natural language generation for patient-facing summaries. Both prompts are executed using the Deepseek-Reasoner LLM and adhere to a constrained vocabulary of ICD-9 terms.

Entity Extraction Prompt The first prompt instructs the model to extract standardized diagnostic entities from unstructured clinical text while enforcing strict adherence to a predefined whitelist of ICD-9 codes. The prompt is organized into structured sections to guide model behavior:

- **Role Definition**: The model is cast as a medical coding assistant restricted to using only terms from a fixed ICD-9 knowledge base (`{knowledge_base}`).
- **Input Format**: The input is framed as a sequence of free-text utterances from a user, denoted by `{text}`.
- **Contextual Entities**: A preliminary list of relevant medical entities (`{entities}`) is provided to focus attention.
- **Task Instructions**: The model is required to:
  1. Assess the severity of each identified entity (in English);
  2. Match entities strictly to the ICD-9 whitelist, allowing for synonym resolution;
  3. Sort the results by descending severity.
- **Output Format**: The output is a structured JSON object of the form:

```
 {
"diagnoses":  [
{ "standard_term":  "...", "match_status":  "Matched/Unknown", "severity":
..., "icd9_code":  "..." },
{ ...  }  ,
, }
```

An illustrative input/output example is embedded within the prompt to ensure consistent generation across samples.

Natural Language Generation Prompt The second prompt reformulates extracted diagnostic codes into natural language summaries appropriate for clinical reporting and patient-facing use. While the core structure mirrors the entity extraction prompt, this prompt emphasizes fluent generation and interpretable phrasing. The key differences include:

```
['Poisoning by barbiturates', 'Poisoning by tetracycline group', 'Other symptoms involving respirato
ry system and chest', 'Palpitations', 'Persistent vomiting']
['9670', '9604', '7869', '7851', '5362']
```

Figure 1: Prompt used for instructing Deepseek-Reasoner to extract structured diagnostic entities from clinical free-text.

```
Diagnosis Output:
Based on your symptom description, our analysis suggests you may be experiencing issues related to t
he following health conditions, listed from strongest to weakest association:

1. **Heart rhythm problems** (Palpitations, ICD9:7851) - Feeling like your heart is racing, skipping
, or pounding
2. **Muscle weakness disorder** (Hereditary progressive muscular dystrophy, ICD9:3591) - Gradual los
s of muscle strength
3. **Tetracycline antibiotic reaction** (Poisoning by tetracycline group, ICD9:35989) - Possible sid
e effects from certain antibiotics
4. **Liver infection** (Amebic liver abscess, ICD9:0063) - Parasitic infection causing liver inflamm
ation
5. **Heart muscle disease** (Other primary cardiomyopathies, ICD9:4254) - Abnormal heart muscle func
tion
```

Figure 2: Example natural language output presenting extracted diagnoses in a fluent, patient-facing summary.

- A temperature setting of 0.1 to promote diverse but coherent expression;
- Emphasis on clarity, empathy, and linguistic smoothness in the output;
- Retention of whitelist constraints to preserve clinical consistency.

The generated output is a ranked list of diagnostic hypotheses expressed in layperson-readable form, again structured as a JSON object for downstream compatibility.

Examples of both prompt types and their generated outputs are shown in Figures 1 and 2.

Generate Patient Description in Natural Language The generated patient description in natural language is show in Figure 3.

C Data

C.1 MIMIC-III Knowledge Graph Processing

We utilize the publicly available MIMIC-III clinical database [1], which comprises de-identified health records from over 40,000 ICU admissions between 2001 and 2012. The dataset includes rich, time-stamped information across multiple domains, including medications, diagnoses, procedures, demographics, and vital signs, making it a robust foundation for modeling patient trajectories and building clinical knowledge graphs.

To convert raw electronic health records (EHRs) into a structured knowledge graph, we extend the preprocessing pipeline introduced in SafeDrug [2]. Our

	diagnoses_icd9	diagnoses_long_title	patient_description
0	['042', '486', '4254', '42820', '4280', '5849'...	['Human immunodeficiency virus [HIV] disease',...	I've been feeling extremely weak and tired all...
1	['99681', '5856', '42833', '41071', '40311', '...	['Complications of transplanted kidney', 'End ...	I've been feeling extremely tired and weak all...
2	['40301', '5856', '53140', '2851', '5609', '42...	['Hypertensive chronic kidney disease, maligna...	I have serious kidney problems that require me...
3	['2766', '5856', '40301', '4254', '4239', '285...	['[UNK:2766]', 'End stage renal disease', 'Hyp...	I've been struggling with multiple health prob...
4	['40301', '58281', '5855', '5990', '4254', '71...	['Hypertensive chronic kidney disease, maligna...	I've been feeling extremely tired and swollen ...
...	...	...	...
95	['5849', '03842', '51881', '5990', '2760', '25...	['Acute kidney failure, unspecified', 'Septice...	I've been feeling really sick since the accide...
96	['51884', '42833', '5849', '2851', '5781', '41...	['Acute and chronic respiratory failure', 'Acu...	I've been struggling to breathe properly and f...
97	['41401', '9971', '42731', '4589', '34680', '3...	['Coronary atherosclerosis of native coronary ...	I often get chest pain and feel like my heart ...
98	['72402', '2761', '2851', '5601', '5070', '276...	['Spinal stenosis, lumbar region, without neur...	I've been having constant pain in my lower bac...
99	['0389', '51884', '73018', '42830', '3241', '4...	['Unspecified septicemia', 'Acute and chronic ...	I've been feeling extremely sick with a high f...

Figure 3: Example of a generated patient description in natural language and the corresponding predicted diagnoses.

	patient_id	visit	diagnoses	procedures	medications	proteins
0	0	1	['4239', '5119', '78551', '4589', '311', '7220...	['3731', '8872', '3893']	['N02B', 'A01A', 'A02B', 'A06A', 'B05C', 'A12A...	[PROTEIN:3897, PROTEIN:6839]
1	0	2	['7455', '45829', 'V1259', '2724']	['3571', '3961', '8872']	['N02B', 'A01A', 'A02B', 'A06A', 'A12A', 'B05C...	[PROTEIN:3897, PROTEIN:6839]
2	1	1	['41071', '78551', '5781', '5849', '40391', '4...	['0066', '3761', '3950', '3606', '0042', '0047...	['A06A', 'B05C', 'C07A', 'A12B', 'C03C', 'A12A...	[PROTEIN:1621, PROTEIN:4405, PROTEIN:4876, PRO...
3	2	1	['2252', '3485', '78039', '4241', '4019', '272...	['0151']	['B05C', 'A07A', 'C07A', 'A06A', 'N02B', 'C02D...	[PROTEIN:3897, PROTEIN:4876, PROTEIN:11410, PR...
4	2	2	['41401', '4111', '4241', 'V4582', '2724', '40...	['3613', '3615', '3961', '8872', '9904', '9905...	['N02B', 'A01A', 'A02B', 'A06A', 'A12A', 'B05C...	[PROTEIN:3897, PROTEIN:6839]

Figure 4: Illustrative subset of the cleaned MIMIC-III data, showing multi-modal alignment of diagnoses, medications, and procedure-derived symptoms for individual visits.

aim is to ensure both temporal consistency and semantic completeness across the three principal clinical modalities: medications, diagnoses, and procedures.

We begin by discarding records lacking valid timestamps or patient identifiers. Only encounters with temporally aligned entries across all three modalities are retained. Medication records are cleaned by removing non-essential details such as dosages, administration routes, and free-text annotations. To reduce data sparsity while preserving clinical diversity, we limit diagnostic and procedural codes to the most frequently occurring entries. Temporal normalization is applied to align visit dates, and missing data is handled via forward imputation within individual patient timelines. Patients with a single visit or incomplete records are excluded to ensure sufficient longitudinal depth.

For improved semantic interpretability, raw clinical codes are mapped to standardized vocabularies. Medications are categorized into higher-level therapeutic groups using the Anatomical Therapeutic Chemical (ATC) classification system. Procedural codes are translated into symptom-related descriptors based on a curated procedure-symptom ontology. In cases where no mapping is available, placeholder tokens are inserted to preserve sequence continuity.

Finally, we merge the cleaned data across all modalities using shared patient and visit identifiers. The resulting dataset comprises multi-modal clinical snapshots, where each visit encodes the patient's diagnostic history, prescribed treat-

ments, and presenting symptoms—providing a rich foundation for graph-based modeling and analysis. An illustrative subset of the cleaned and structured cohort is visualized in Figure 4.

C.2 Oregano Knowledge Graph

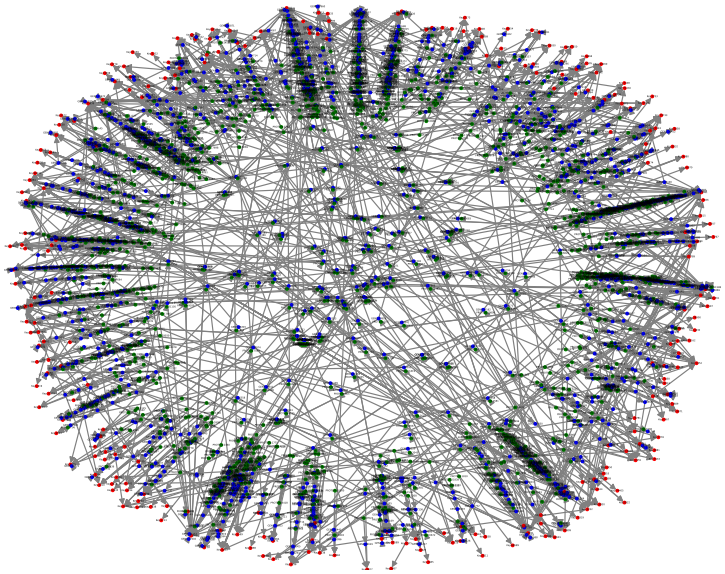

Figure 5: Overview of the Oregano Biomedical Knowledge Graph composed of curated entities and relations across clinical ontologies.

We construct the Oregano Knowledge Graph by following the official pipeline provided in the Awesome Biomedical Knowledge Graphs project (Haas et al., 2023)[1]. This pipeline integrates multiple biomedical ontologies and curated databases to generate a heterogeneous graph that supports reasoning over clinical and molecular concepts.

The graph incorporates diverse sources such as UMLS (Unified Medical Language System), DrugBank, MeSH, and SNOMED CT. Entities in the graph represent biomedical concepts—including diseases, drugs, procedures, and symptoms—while edges represent semantically typed relations such as `treats`, `causes`, `has_symptom`, and `interacts_with`.

To generate the graph, we clone the public repository and run the extraction and normalization scripts. Entity alignment is performed using UMLS Concept Unique Identifiers (CUIs) to ensure semantic consistency across vocabularies. The final graph is serialized in structured formats (e.g., CSV triples or RDF), with each triplet storing a `head` entity, `relation` type, and `tail` entity.

[1] Available at: Robert Haas et al., Awesome Biomedical Knowledge Graphs, 2023.

Figure 5 presents an overview of the Oregano Knowledge Graph, which serves as a core biomedical knowledge base for downstream tasks such as diagnostic reasoning, phenotype prediction, and drug repurposing.

D Knowledge Reasoning Rule

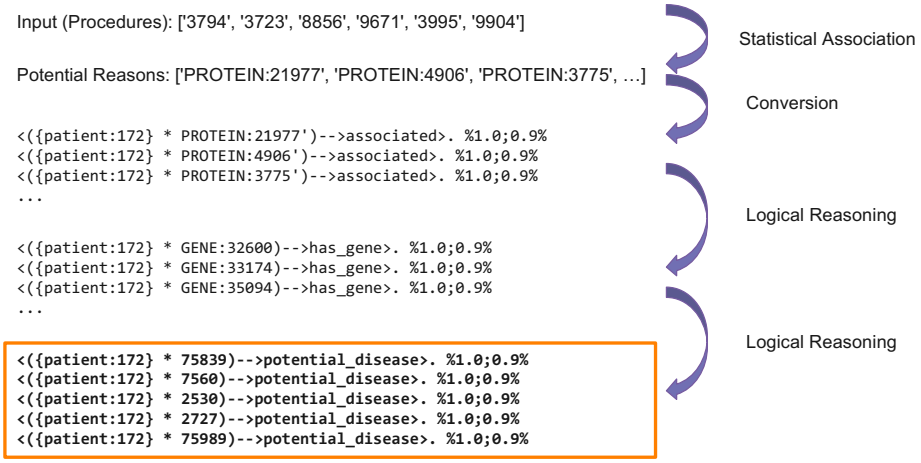

Figure 6: Illustration of diagnostic reasoning over the Oregano Knowledge Graph. The input is mapped to candidate rationales and converted into *Narsese* knowledge representations. The reasoning engine then performs two steps of logical inference to derive new facts, ultimately identifying a potential disease (bold).

E Results

E.1 ReCLLaMA Interface Overview

We present the user interface of the ReCLLaMA system in Figure 10. The interface is designed to support real-time, multi-turn interactions with users via natural language. It allows patients or clinicians to pose clinical questions and receive structured, interpretable diagnostic feedback. The system maintains session state across turns, enabling context-aware dialogue and follow-up queries (e.g., "Explain how Gene 4254 affects heart function"). Supporting evidence—such as gene-disease associations, protein annotations, and mechanistic descriptions—is embedded in expandable sections for traceability and interpretability.

- Unspecified septicemia
- Pneumonitis due to inhalation of food or vomitus
- Acute on chronic systolic heart failure
- Congestive heart failure, unspecified
- Acute respiratory failure
- Subendocardial infarction, initial episode
- Atrial fibrillation
- Severe sepsis
- Diabetes mellitus without mention of complication, type II or unspecified type, not stated as uncontrolled
- Other persistent mental disorders due to conditions classified elsewhere
- Pure hypercholesterolemia
- Old myocardial infarction
- Esophageal reflux
- Long-term (current) use of anticoagulants
- Unspecified essential hypertension
- Fitting and adjustment of artificial leg
- Fitting and adjustment of other specified prosthetic device

Figure 7: Gold reference explanation manually authored by clinical experts.

E.2 Diagnostic Response Comparison

To evaluate ReCLLaMA's diagnostic reasoning capabilities, we conduct a comparative study using a real-world test case introduced in the last part of Result section. We benchmark the system against ground truth expert annotations and ChatGPT-4.0's generative outputs. Figures 7, 8, and 9 provide visual comparisons of diagnostic explanations generated by each source:

- Figure 7 shows the reference explanation curated by human medical experts.
- Figure 8 displays ReCLLaMA's generated explanation, grounded in symbolic reasoning and supported by biomedical knowledge graphs.
- Figure 9 illustrates the generative output from ChatGPT-4.0 for the same input query.

The comparison reveals that ReCLLaMA produces responses that are more

AI-generated explanation:

Explanation for the Patient

Possible Health Conditions Based on Your Symptoms:

1. **Congenital Musculoskeletal Anomalies** (e.g., birth differences in bones/joints).
2. **Neuromuscular Disorders** (e.g., nerve-muscle communication issues causing weakness or cramps).
3. **Congenital Esophageal Issue** (e.g., abnormal widening of the food pipe).
4. **Kidney Development Issue** (e.g., missing or underdeveloped kidneys).
5. **Thyroid Cancer** (abnormal growth in the thyroid gland).

These conditions are flagged with **high confidence** (90%) in the analysis.

If You Want to Know "Why These Symptoms Occur":

These symptoms may be linked to **genetic factors**. Genes act like "instruction manuals" for your body. When certain genes don't work properly, they can affect how muscles, nerves, or organs develop. For example:

- **GENE:34156**: May influence muscle and nerve function.
- **GENE:30233**: Could play a role in kidney development.
- **GENE:32294**: Might affect the thyroid gland.
- **GENE:32922**: Possibly linked to connective tissue or bone health.

(Note: These gene codes are simplified for clarity. A genetic counselor can provide detailed insights.)

Figure 8: ReCLLaMA-generated explanation produced via symbolic reasoning over structured biomedical knowledge graphs.

aligned with expert knowledge and traceable to biomedical sources, while ChatGPT-4.0 provides fluent but less grounded outputs.

E.3 System Demonstration

To complement the screenshots, we also provide a video demo that illustrates ReCLLaMA's real-time interaction on the same test case. The demo showcases the system's ability to extract clinical entities, perform inference over the knowledge graph, and generate structured diagnostic responses with supporting evidence. The demonstration is available as supplementary material.

References

[1] A. E. Johnson, T. J. Pollard, L. Shen, L.-w. H. Lehman, M. Feng, M. Ghassemi, B. Moody, P. Szolovits, L. Anthony Celi, and R. G. Mark, "Mimic-iii,

Based on your detailed symptoms, here's an AI-generated medical explanation integrating potential diagnoses and related biological mechanisms:

Likely Diagnoses (based on ICD-9 symptom codes and clinical interpretation):

1. **Photosensitivity Reaction**
   - **ICD-9:** 692.72 — Contact dermatitis due to solar radiation
   - Your description of severe skin blistering and redness upon sunlight exposure is characteristic of photosensitivity, which can be **drug-induced** (especially antibiotics or sedatives).
2. **Drug-Induced Toxicity / Overdose Syndrome**
   - **ICD-9:** 969.4 — Poisoning by benzodiazepines and other sedative-hypnotics
   - Sedation, confusion, muscle weakness, and cardiovascular effects like racing/fluttering heart suggest CNS depressant effects.
3. **Gastrointestinal Distress / Acute Gastritis**
   - **ICD-9:** 535.4 — Other specified gastritis with hemorrhage
   - Severe cramping, vomiting bile, and doubled-over pain point to acute irritation of the gastrointestinal lining.
4. **Autonomic Dysregulation / Panic-like Toxic Response**
   - **ICD-9:** 780.2 — Syncope and collapse
   - The sensation of gasping for air, rapid puls[illegible]nd blurred vision may result from systemic

Figure 9: ChatGPT-4.0 generated explanation without access to structured biomedical knowledge or symbolic inference.

a freely accessible critical care database," *Scientific data*, vol. 3, no. 1, pp. 1–9, 2016.

[2] C. Yang, C. Xiao, F. Ma, L. Glass, and J. Sun, "Safedrug: Dual molecular graph encoders for safe drug recommendations," in *Proceedings of the Thirtieth International Joint Conference on Artificial Intelligence, IJCAI 2021*, 2021.

Figure 10: ReCLLaMA user interface for diagnostic question answering, supporting structured outputs and evidence-grounded explanations.